

SCALE-COLLAPSE COST EXCLUSION FOR TYPE II NAVIER-STOKES RESCALINGS

GUILLEM CORDERO

ABSTRACT. We analyze Type II Navier–Stokes rescalings for which the renormalized scale collapses along the tail. In a scale-translation gauge, the logarithmic scale satisfies $\partial_\tau \log \lambda = a$. If the collapsing rescaling has a nonzero localized L^2 core, then the negative part of the scale drift has infinite localized accumulation. This infinite accumulation is itself the scale-collapse Type II cost. Consequently finite-cost Type II branches cannot realize genuine scale collapse with a nonvanishing localized core. An absolute scale-drift cost gives the same conclusion by pointwise domination. We also record the autonomous stationary-profile reduction as a secondary rigidity route; it is not needed for the finite-cost scale-collapse exclusion.

1. RENORMALIZED SCALE COLLAPSE

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

near a putative Type II singular time. We consider a terminal rescaling written in a scale-translation gauge:

$$u(x, t) = \lambda(\tau)^{-1} V \left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right), \quad p(x, t) = \lambda(\tau)^{-2} P \left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right) + c(t),$$

where $\lambda(\tau) > 0$. The corresponding renormalized equation is

$$(1) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

The scale convention is

$$(2) \quad \frac{d}{d\tau} \log \lambda(\tau) = a(\tau).$$

Thus physical scale collapse $\lambda(\tau) \rightarrow 0$ is associated with negative logarithmic scale drift.

Let $\Phi(\tau, \cdot) \in C_c^\infty(\mathbb{R}^3)$ be a nonnegative moving cutoff adapted to the local scale-collapse core, and define

$$M_R(\tau) = \int_{\mathbb{R}^3} |V(y, \tau)|^2 \Phi(\tau, y) dy.$$

A nonvanishing localized core means that there are $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \geq \tau_1.$$

Definition 1.1 (Scale-collapse and absolute scale costs). The scale-collapse cost density is

$$\mathfrak{C}_{\text{sc}}(\tau) = a_-(\tau) M_R(\tau), \quad a_-(\tau) = \max\{-a(\tau), 0\}.$$

The absolute scale-drift cost density is

$$\mathfrak{C}_{\text{abs}}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 \Phi(\tau, y) dy + |a(\tau)| M_R(\tau).$$

The corresponding cumulative costs on $[\tau_0, \infty)$ are the integrals of these nonnegative densities.

Definition 1.2 (Finite-cost scale-collapse branch). A scale-collapse branch is finite-cost in the scale-collapse sense if

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{sc}}(\tau) d\tau < \infty.$$

It is finite-cost in the absolute scale-drift sense if

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{abs}}(\tau) d\tau < \infty.$$

2. SCALE-DRIFT DIVERGENCE

Lemma 2.1 (Finite-or-infinite alternative). *For every measurable renormalized solution for which $a_- M_R$ is measurable and nonnegative, exactly one of the following alternatives holds:*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau < \infty, \quad \text{or} \quad \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau = \infty.$$

Proof. The integral is an extended nonnegative number. Hence it is either finite or infinite, and the two cases are mutually exclusive. $\square$

Lemma 2.2 (Logarithmic scale identity). *Under the convention (2), for every $\tau_0 \leq \tau_1 < \tau_2 < \infty$,*

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Proof. This is the fundamental theorem of calculus applied to $\partial_\tau \log \lambda = a$. $\square$

Theorem 2.3 (Collapse with a nonvanishing core forces infinite scale-collapse cost). *If*

$$\lambda(\tau) \rightarrow 0 \quad \text{as } \tau \rightarrow \infty,$$

and the rescaling has a nonvanishing localized core, then

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau = \int_{\tau_0}^{\infty} \mathfrak{C}_{\text{sc}}(\tau) d\tau = \infty.$$

Proof. By Theorem 2.2,

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Since $\lambda(\tau) \rightarrow 0$, the right-hand side tends to $-\infty$ as $\tau_2 \rightarrow \infty$. If $\int_{\tau_1}^{\infty} a_-(\tau) d\tau < \infty$, then

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau) d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau) d\tau \geq - \int_{\tau_1}^{\infty} a_-(\tau) d\tau > -\infty,$$

contradicting the divergence to $-\infty$. Thus $\int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty$.

The core floor gives $M_R(\tau) \geq m_0$ for a.e. $\tau \geq \tau_1$. Hence

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau \geq m_0 \int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty.$$

$\square$

3. FINITE-COST EXCLUSION

Theorem 3.1 (Scale-collapse cost exclusion). *No finite-cost scale-collapse branch in the sense of Theorem 1.2 can satisfy both $\lambda(\tau) \rightarrow 0$ and the nonvanishing localized-core condition.*

Proof. By Theorem 2.3, genuine scale collapse with a nonvanishing localized core gives

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{sc}}(\tau) d\tau = \infty.$$

This contradicts finite cost in the scale-collapse sense. $\square$

Lemma 3.2 (Absolute cost domination). *For every τ ,*

$$\mathfrak{C}_{\text{abs}}(\tau) \geq \mathfrak{C}_{\text{sc}}(\tau).$$

Consequently,

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{sc}}(\tau) d\tau = \infty \implies \int_{\tau_0}^{\infty} \mathfrak{C}_{\text{abs}}(\tau) d\tau = \infty.$$

Proof. Pointwise,

$$\mathfrak{C}_{\text{abs}}(\tau) = \nu \int |\nabla V|^2 \Phi + |a(\tau)| M_R(\tau) \geq |a(\tau)| M_R(\tau) \geq a_-(\tau) M_R(\tau) = \mathfrak{C}_{\text{sc}}(\tau).$$

Integrating gives the implication. $\square$

Corollary 3.3 (Absolute scale-cost exclusion). *No finite-cost scale-collapse branch in the absolute scale-drift sense can satisfy both $\lambda(\tau) \rightarrow 0$ and the nonvanishing localized-core condition.*

Proof. Combine Theorems 2.3 and 3.2. The resulting infinite absolute scale-drift cost contradicts finite cost in the absolute scale-drift sense. $\square$

4. AUTONOMOUS RIGIDITY ROUTE

The cost route above is the scale-collapse exclusion used in the finite-cost Type II analysis. There is also a secondary PDE rigidity route. It is useful when one wants to identify the scale-collapse obstruction with a stationary or generalized self-similar profile. This route is not needed for Theorems 3.1 and 3.3.

Definition 4.1 (Thick autonomous scale-collapse windows). A scale-collapse rescaling has thick autonomous windows if there are $T > 0$, $c > 0$, $\tau_n \rightarrow \infty$, constants $a_\infty \in \mathbb{R}$, $b_\infty \in \mathbb{R}^3$, and a constant $M_1 < \infty$ such that

$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau) d\tau \geq c, \quad M_R(\tau) \leq M_1$$

for a.e. $\tau \in [\tau_n, \tau_n + T]$, and

$$a(\tau_n + \cdot) \rightarrow a_\infty, \quad b(\tau_n + \cdot) \rightarrow b_\infty$$

in a coefficient topology sufficient to pass the coefficient terms in distributions. Strong convergence in $L^1(0, T)$ is sufficient.

Lemma 4.2 (Thick drift gives a negative autonomous scale parameter). *On thick autonomous scale-collapse windows,*

$$a_\infty \leq -c/M_1 < 0.$$

Proof. The upper mass bound and thick weighted drift give

$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) d\tau \geq c/M_1.$$

After translation and $L^1(0, T)$ convergence of a to a_∞ , the left-hand side converges to $(-a_\infty)_+$. Therefore $(-a_\infty)_+ \geq c/M_1$, which gives the claim. $\square$

Theorem 4.3 (Autonomous reduced-limit route). *Consider translated windows*

$$V_n(y, s) = V(y, \tau_n + s), \quad P_n(y, s) = P(y, \tau_n + s), \quad 0 < s < T.$$

If, on thick autonomous scale-collapse windows, the sequence (V_n, P_n) is locally compact strongly enough to pass the nonlinear and pressure terms and the localized core floor passes to a compact subcutoff in the limit, then after passing to a subsequence (V_n, P_n) converges to a nonzero weak solution (U, Π) of

$$(3) \quad \partial_s U + (U \cdot \nabla)U + \nabla \Pi = \nu \Delta U + a_\infty(U + y \cdot \nabla U) + b_\infty \cdot \nabla U, \quad \nabla \cdot U = 0.$$

Proof. The local $L^2 H^1$ bound and time compactness give, by Aubin–Lions, strong local convergence in $L^2(0, T; L^q(B_R))$ for every $q < 6$. This strong local convergence, together with weak $L^2 H^1$ convergence, passes the nonlinear term to $(U \cdot \nabla)U$ in distributions. The pressure convergence passes the Poisson reconstruction to Π . The modulation-limit convergence passes the coefficient terms to $a_\infty(U + y \cdot \nabla U)$ and $b_\infty \cdot \nabla U$. The localized L^2 floor passes to the limit on a compact subcutoff, so $U \not\equiv 0$. $\square$

Theorem 4.4 (Stationary-profile rigidity route). *If the autonomous limit in Theorem 4.3 has a nonzero stationary omega-limit (W, Q) in a class where the corresponding stationary or generalized self-similar Liouville theorem applies, then the scale-collapse branch is excluded by that rigidity theorem.*

Proof. The stationary omega-limit satisfies

$$(W \cdot \nabla)W + \nabla Q = \nu \Delta W + a_\infty(W + y \cdot \nabla W) + b_\infty \cdot \nabla W, \quad \nabla \cdot W = 0.$$

It is nonzero by construction. The applicable Liouville or rigidity theorem rules out such a nonzero admissible stationary profile, giving a contradiction. $\square$

Remark 4.5 (Role of the autonomous route). The autonomous route is a structural description of a special scale-collapse subcase. Infinite scale-drift cost alone does not force thick windows, coefficient convergence, a stationary omega-limit, or a parameter-specific Liouville theorem. Those conditions are not needed for the finite-cost scale-collapse exclusion, which follows directly from Theorems 3.1 and 3.3.

REFERENCES

- [NRS96] J. Necas, M. Ruzicka, and V. Sverak, On Leray’s self-similar solutions of the Navier–Stokes equations, *Acta Math.* **176** (1996), 283–294.